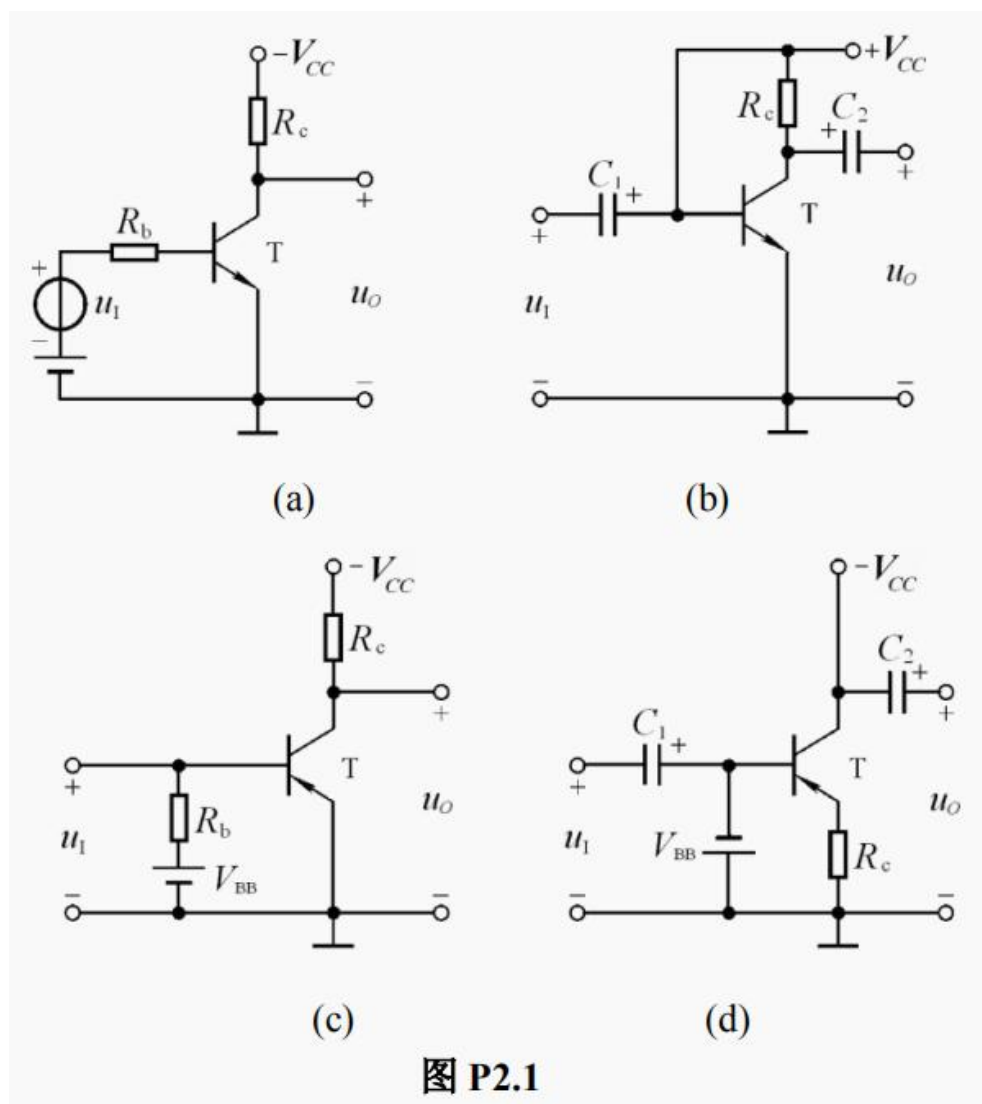


Homework 2

- ✧ 动态和静态分析: 教材 2.1/2.2/2.4 (共 3 题)
- ✧ 放大电路参数计算: 教材 2.7/2.9/2.10/2.11 (共 4 题)

2.1 分别改正图 P2.1 所示各电路中的错误,使它们有可能放大正弦波信号。要求保持电路原来的共射接法

2.1 The circuits below are not connected properly. Correct the errors in each circuit shown in figure p2.1 respectively, so that they may amplify the sine wave signal. Please don't change the type of amplifier circuits (For example, if it is common-emitter circuit, after correction it should also be common-emitter circuit).



2.2 画出图 P2.2 所示各电路的直流通路和交流通路。设所有电容对交流信号均可视为短路。

2.2 Draw the DC and AC circuits of each amplifier circuit as shown in figure P2.2. Assume all capacitors have no resistance to AC signals.

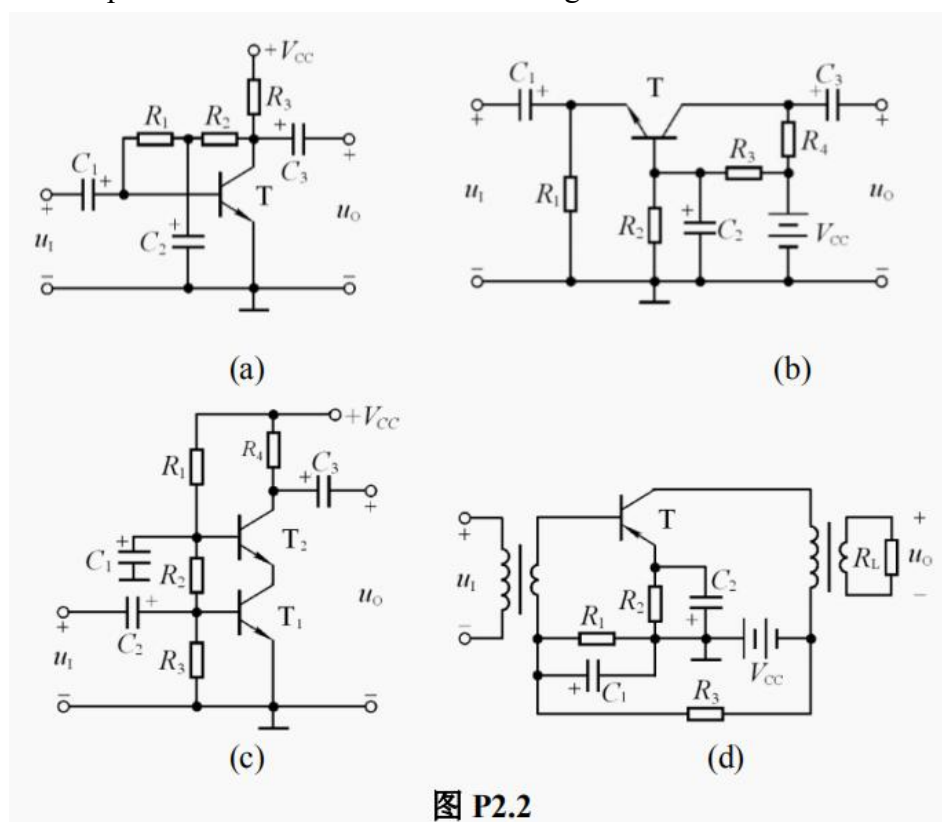


图 P2.2

2.4 电路如图 P2.4(a)所示,图(b)是晶体管的输出特性,静态时 $U_{BEQ}=0.7V$ 。利用图解法分别求出 $R_L=\infty$ 和 $R_L=3\text{ k}\Omega$ 时的静态工作点和最大不失真输出电压 U_{OM} (有效值) (有效值=峰峰值/ $2\sqrt{2}$)。

2.4 Figure P2.4 (a) shows an amplifier circuit. Figure (b) shows the output characteristics of the transistor. The quiescent voltage on BE junction is $U_{BEQ}=0.7V$. Ask: (1) Calculate the quiescent working point and the maximum undistorted output voltage U_{OM} (effective value) at $R_L = \infty$ using the graphic method; (2) Calculate the quiescent working point and the maximum undistorted output voltage U_{OM} (effective value) at $R_L = 3K\ \Omega$ using the graphic method.

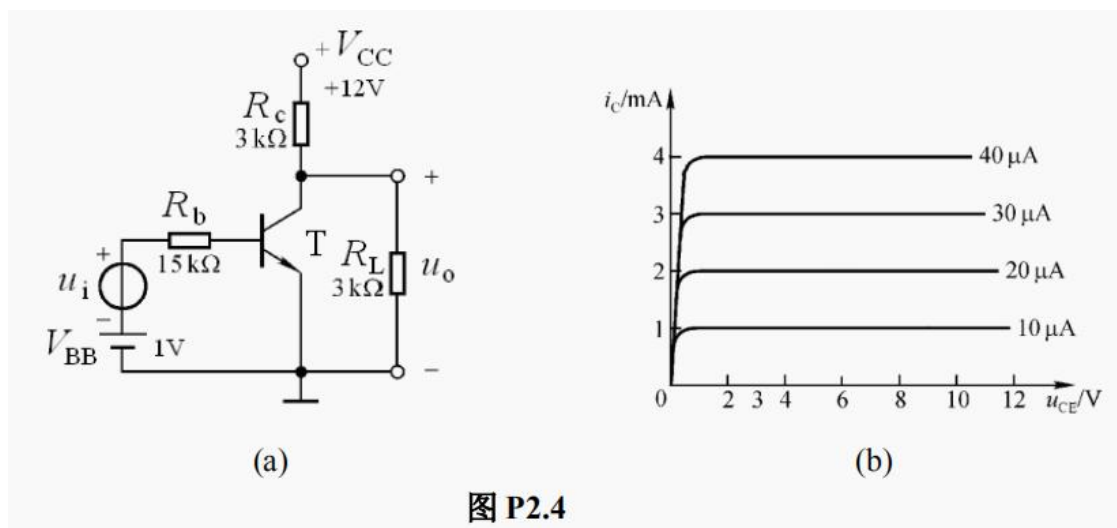


图 P2.4

2.7 电路如图 P2.7 所示,晶体管的 $\beta=80$, $r_{bb'}=100\Omega$ 。分别计算 $R_L=\infty$ 和 $R_L=3k\Omega$ 时的静态工作点 Q 点 (I_{BQ} , I_{CQ} , U_{CEQ})、电压放大倍数 A_u 、输入电阻 R_i 和输出电阻 R_o 。

2.7 An amplifier circuit is shown in figure P2.7. Assume $\beta=80$, $r_{bb'}=100\Omega$, Ask:

- (1) When $R_L=\infty$, find the quiescent working point (I_{BQ} , I_{CQ} , U_{CEQ}), AC voltage gain A_u , input resistance R_i , and output resistance R_o ;
- (2) When $R_L=3k\Omega$, find the quiescent working point (I_{BQ} , I_{CQ} , U_{CEQ}), AC voltage gain A_u , input resistance R_i , and output resistance R_o .

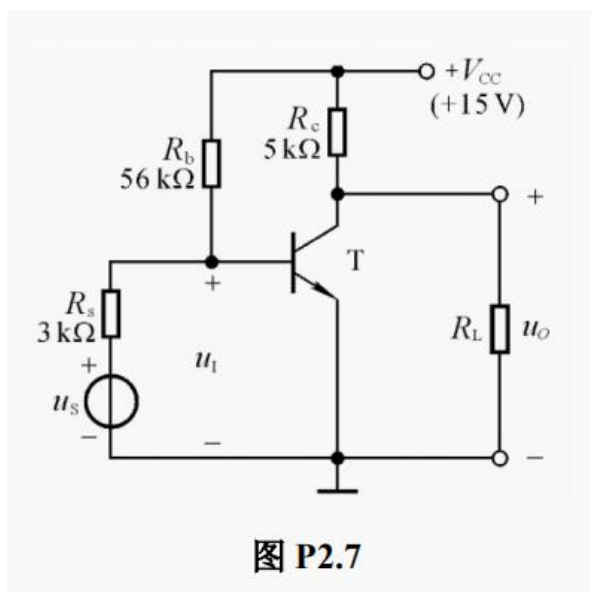


图 P2.7

2.9 已知图 P2.9 所示电路中晶体管的 $\beta=80$, $r_{be}=1.4k\Omega$

- (1) 现已测得静态管压降 $U_{CEQ}=6V$,估算 R_b 约为多少千欧?
- (2) 若测得 u_i 和 u_o 的有效值分别为 $1mV$ 和 $100mV$,则负载电阻 R_L 约为多少千欧?

2.9 For the amplifier circuit shown in figure p2.9, $\beta=80$ and $r_{be}=1.4k\Omega$

- (1) What's the value of R_b when $U_{CEQ}=6V$
- (2) When $u_i=1mv$, $u_o=100mV$, what's the value of R_L =?

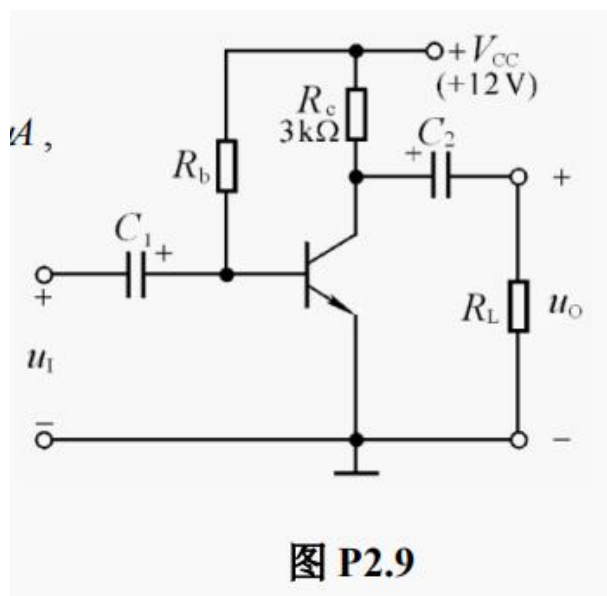


图 P2.9

2.10 在图 P2.9 所示电路中,设静态时 $I_{CQ}=2\text{mA}$, 晶体管饱和管压降 $U_{CES}=0.6\text{V}$ 。试问: 当负载电阻 $R_L=\infty$ 和 $R_L=3\text{k}\Omega$ 时电路的最大不失真输出电压各为多少伏?

2.10 The amplifier circuit is shown in Figure P2.9. At quiescent working points $I_{CQ}=2\text{mA}$ and the voltage drop on CE junction when enters into saturation region is $U_{CES}=0.6\text{V}$. Ask:

- (1) When the load resistance $R_L = \infty$, what is the maximum undistorted output voltage of the circuit (dynamic range)?
- (2) When the load resistance $R_L = 3\text{k}\Omega$, what is the maximum undistorted output voltage of the circuit (dynamic range)?

2.11 电路如图 P2.11 所示,晶体管的 $\beta=100$, $r_{bb'}=100\Omega$ 。

- (1) 求电路的 Q 点、电压放大倍数 A_u 、输入电阻 R_i 和输出电阻 R_o ;
- (2) 若改用 $\beta=200$ 的晶体管,则 Q 点如何变化?
- (3) 若电容 C_e 开路,则将引起电路的哪些动态参数发生变化?如何变化?

2.11 The amplifier circuit is shown in figure p2.11. The transistor $\beta=100$, $r_{bb'}=100\Omega$.

- (1) Find quiescent working point Q (I_{BQ} , I_{CQ} , U_{CEQ}), AC voltage gain A_u , input resistance R_i , and output resistance R_o ;
- (2) If the transistor is replaced with another one with $\beta = 200$, how does the Q point change?
- (3) If we remove capacitor C_e , how will dynamic parameters (A_u , R_i , and R_o) change?

